

KOLLURU VANI

Embedded Software Engineer | AUTOSAR | Automotive Embedded Systems

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PROFESSIONAL SUMMARY

Highly skilled **Embedded Software Engineer** with **4+ years of experience** in the automotive domain, specializing in **AUTOSAR-based ECU software development, integration, and validation**. Proven track record of delivering complex projects for **Tata Sierra (EV & ICE)** and **Volkswagen MPCl Gateway ECU** platforms. Expert in **Embedded C, CAN, CANoe, and UDS Diagnostics** with a strong focus on software quality and **ASPICE-compliant processes**. Recognized for excellence through Employee of the Month and company-wide recognition.

CORE COMPETENCIES

AUTOSAR Standards: AUTOSAR Architecture, SWC Development, MCAL Configuration, BSW Integration, RTE
Programming & Languages: Embedded C, C Programming
ECU Development & Testing: ECU Software Development, Software Integration, Unit Testing (Cantata), Integration Testing, MC/DC Coverage, Functional & Robustness Testing
Communication Protocols: CAN, SPI, I2C, UART
Debugging & Validation: Trace32, UDE, CANoe, UDS Diagnostics, Root Cause Analysis, Software Validation
Development Tools & OS: Jira, Git, Bitbucket, DOORS, Rhapsody, Enterprise Architect, Windows, Linux
Domain Expertise: Automotive Lighting & Sensing Systems, Gateway ECUs, Autonomous Driving Platforms (L4)
Process: ASPICE-based Development Lifecycle

PROFESSIONAL EXPERIENCE

Marelli India Pvt. Ltd. | Software Engineer | May 2024 - Present

Project: Tata Sierra (EV & ICE) – Automotive Lighting & Sensing Systems

- Spearheaded AUTOSAR-based ECU software development for advanced automotive lighting systems including DRL, High Beam, Low Beam, and Turn Indicators.
- Developed and validated AUTOSAR Software Components (SWCs) and implemented complex animation sequences for front headlamps.
- Executed rigorous unit and integration testing, debugging, and root cause analysis to ensure software reliability and performance.
- Provided critical onsite customer support at Tata Motors, Pune, serving as a key technical interface during vehicle integration and validation.
- Successfully resolved critical front headlamp animation issues, ensuring key project milestones were achieved.
- Collaborated with System, Hardware, and Validation teams across the software development lifecycle.

Bosch Global Software Technologies | Associate Software Engineer | July 2022 - May 2024

Project: Volkswagen MPCl Gateway ECU – Autonomous Driving Platform

- Contributed to the development of a Gateway ECU for Level-4 autonomous driving vehicles.
- Performed MCAL configuration, software integration, and state machine development.
- Developed Software Unit Test Plans (SUPs) and executed Cantata-based testing with MC/DC coverage.
- Conducted functional, robustness, and coverage testing while managing defect resolution using Trace32 and UDE.
- Supported software flashing, remote testing, and validation in ASPICE-based development environments.

Vector India | Embedded Systems Trainee | Dec 2021 - Jul 2022

- Completed hands-on training in C, C++, and Embedded C programming for microcontroller-based systems.
- Gained practical knowledge of 8051 microcontroller architecture, memory organization, and peripheral interfacing.
- Developed mini-projects utilizing UART, SPI, and I2C communication protocols.

EDUCATION

Bachelor of Technology (B.Tech) in Electronics & Communication Engineering

Mother Theresa College of Engineering & Technology | June 2017 - July 2021

CERTIFICATIONS

Embedded Systems Training – Vector India, Hyderabad

AWARDS & RECOGNITION

- **Employee of the Month (September 2025)** – Awarded for exemplary performance and rapid resolution of critical software issues at Marelli India.
- **Featured in Quarterly Newsletter (Q4 2025)** – Recognized for innovation, adaptability, and collaboration.
- **Onsite Customer Support Recognition** – Appreciated for successful support and contribution to Tata Sierra integration at Tata Motors, Pune.